

Work orders: executing the CZB roadmap with a smaller model

2026-08-27. Jonas asked whether the roadmap's execution can move to a less expensive model, and what would facilitate that. This document is the facilitation: task cards precise enough that a capable but cheaper session can execute them without design judgment, with acceptance criteria that make success checkable and stop conditions that route genuine judgment calls back to Jonas or a stronger review session. The recommendation on when to use which model is in section 1; the cards follow.

1 Recommendation on the model split

My honest view: most of the remaining work is well-specified implementation against existing patterns, and a cheaper model executes that reliably *if* the specification carries the judgment. The split I would use:

- **Cheaper model, per these cards:** A.1, A.2, A.4, B.1–B.3 constructions, C, E.1, document rebuilds, and all reruns.
- **Stronger model (or Jonas):** interpreting A.2's bias-model comparison, A.3's pass/fail verdict, B.4's transfer readout, anything where a card's stop condition fires, and a periodic review pass over completed cards (the pattern that caught the paired-difference overstatement is worth keeping: execute cheap, review strong).
- Do not hand a cheaper session an open question ("does this look right?") — hand it a card. When a card's acceptance criterion cannot be met, the correct output is a short failure report, not an improvisation.

2 Standing rules for every card (read first, they are load conditions)

- 1 **Every number that will be quoted anywhere must come from a committed script** with its output file tracked. No in-session-only analyses — this rule exists because an uncommitted analysis produced an overstated headline once already (results doc §4.3b-ii, corrected 2026-08-27).
- 2 **Run** `python tests/test_surprise.py, test_comfortzone.py, test_causation.py, test_cutin.py` **before and after each card**; all must pass. New behavior gets new property tests in the same `check()` style.
- 3 **Never modify** `src/aidriver/preferences.py` **defaults** — released behavior is frozen; new behavior goes behind flags defaulting off.
- 4 **House style** for anything written for Jonas: US English, markdown as source, Word + PDF generated (`pandoc + docs/build_pdf.py`), no period ending a heading, opinions hedged as opinions. OneDrive locks: if a build fails with `PermissionError`, write a -vN name and say so.
- 5 **Environment:** Python 3.14, torch CPU-only, no LaTeX/GPU/gh. Long jobs checkpointed and restartable; background runners never piped through `tail`.
- 6 **Provenance discipline:** cite only verified references; mark anything unverified.

3 Task cards

Card S.1 — draft the `perform-research` skill (first task for the next session)

- **Goal:** a reusable, iteratively updatable skill that captures how research is performed in this collaboration — so the working discipline survives model changes, session changes, and time. Jonas asked for it explicitly, drafting delegated to the next (cheaper) session, with the instruction to also look at what exists elsewhere online and borrow with attribution.
- **Build:** draft `C:\Users\bargman\claude\skills\perform-research\SKILL.md` with frontmatter (name, description triggering on research execution, analysis tasks, fitting runs, and model handovers in this project family). Required contents, from this project's accumulated practice:
- **Reproducibility discipline:** every quoted number from a committed script with tracked output; regeneration commands logged; generated artifacts never hand-edited; the 2026-08-27 paired-difference correction as the cautionary tale.
- **Verification discipline:** property tests before and after every task; verify by two independent routes where feasible; results that contradict documented findings get surfaced, never silently reconciled.
- **Documentation discipline:** markdown as source of truth, Word/PDF generated; dated handover files per arc plus a standing handover `.md` entry point; decisions recorded with their reasoning and their location; the `jonas-academic-writing` conventions for anything addressed to Jonas.

- **Between-model handover protocol:** the entry sequence (handover chain → roadmap → work orders), one card per session, the escalation triggers and review gates (copy from handover_2026-08-27.md §8 — that list is the current authority), worklog appends, and the rule that a cheaper session executes cards and never improvises around a failed acceptance criterion.
- **Statistical practice:** uncertainty conventions stated and resampled completely (all variance sources); comparisons as differences; held-out validation for any model comparison; pre-stated decision rules for verdict experiments.
- **Research step:** search online for existing practice worth borrowing — multi-agent / multi-model handoff patterns, LLM-assisted research workflow guides, lab-notebook and reproducibility standards (e.g. from open-science communities), and any published skill/prompt libraries for research execution. Cite what is borrowed; mark unverified claims per house rules.
- **Accept:** the draft SKILL.md exists, is under ~300 lines, cites its sources, and ends with an explicit "for Jonas's review" note. **Do not treat the skill as active until Jonas approves it;** iterate on his comments in later sessions.
- **Stop if:** unsure whether some practice is project law or one session's habit — list the uncertain items in the draft for Jonas to rule on rather than guessing.

Card R.1 — review gate after A.2/A.3/A.4 (for the stronger model)

[Executed 2026-08-27 by a tier-1 review session. Decisions: hierarchical lapse is the primary A.2 variant (evidence: replication/czb/out/bias_variant_diagnostics.md); the A.3 accumulator is repaired by gating accumulation at manoeuvre onset, and A.3/A.4 were re-run at the gate as v2. All nine queries resolved or carried in the work log; register regenerated. Follow-up work is card A.2.v2 below.]

- **Goal:** close the queries the executing session raised, in severity order, and decide the two things it deliberately did not decide.
- **Read first:** replication/czb/out/query_register.md (generated — the work log is the source of truth), then stage1_summary.md, stage2_summary.md, button_validation_summary.md, then the three worklog entries dated 2026-08-27.
- **The two decisions that block downstream work:**
- **A.2.Q2 — the bias variant.** Group-level versus hierarchical lapse moves the between-driver spread 0.341 → 0.209, a 39% change in the quantity a percentile is made of, while held-out likelihood separates them by 0.6 units over 3 096 trials. The fitted per-driver lapse spread is large (sd 2.76, logit scale), so the hierarchical lapse absorbs variation the group model gave to the threshold — the opposite direction to the leakage argument in docs/czb_validation_roadmap.md §5.1, which should be revisited or corrected in the light of it. Deciding this fixes the headline percentile. *[Corrected at the gate, 2026-08-27: the claimed contradiction was a misreading. Roadmap §5.1 predicted that under real leakage the GROUP model shows the wider σ_c — which is exactly the observed ordering (0.341 > 0.209), so the observation confirms §5.1 rather than contradicting it. Decision and evidence: replication/czb/out/bias_variant_diagnostics.md.]*
- **A.3.Q1 — the accumulator's specification.** The pre-registered verdict is FAIL, but the in-sample fit is worse than the simpler model it extends, and the cause is identified: noise accumulating through 15.1 s of empty pre-onset clip. Choose the remedy (leaky accumulator; accumulation starting where evidence exists; or something else), then A.3 and A.4 are re-run and their verdicts mean something. Until then neither FAIL should be quoted as evidence about the framework.
- **Also verify, not just accept:** that the numbers quoted in the three summaries come from the committed scripts (the failure mode with precedent here is a number that exists only in a transcript); and that A.2's percentile table is read with its two minor caveats attached (A.2.Q4, A.2.Q5).
- **Owed work the executing session named:** A.2.Q3 asks for the pre-onset predictive under *both* bias variants; only the better-fitting one was checked.
- **Accept:** every open query either resolved with a RESOLVED <card>.Q<n>: . . . line appended to the work log, or explicitly carried forward with a reason; the register regenerated; and if a decision changes a documented plan, that document corrected in place with a dated note.

Card A.2.v2 — regenerate stage 1 under the corrected validation code, and test the correlated-effects variant

(Added at review gate R.1, 2026-08-27. Cheap-model card; long-running, overnight is fine.)

- **Why:** R.1 found and fixed two defects in `fit_stage1.py`'s validation code (not in the fits): the hierarchical LOPO integrated the two random effects on the same quadrature nodes (the diagonal of the 2D integral, asserting perfect rank correlation), and the C1 predictive plugged in the hier variant's *median* lapse where the population mean is required. The stored `out/stage1_summary.md` therefore carries a LOPO comparison and a C1 table computed under those defects. R.1 also found the fitted `b_i` and `c_i` correlate at Spearman -0.700 while the model assumes independence, and specified a correlated-effects check.
- **Build:** (a) re-run `python replication/czb/fit_stage1.py` as committed (the fixes are in place; 15 LOPO folds as before). (b) Add `fit_hier_corr` to `fit_stage1.py`: the hierarchical-lapse model with correlated effects — driver threshold uses $z1$, driver lapse uses $p \cdot z1 + \sqrt{(1-p^2)} \cdot z2$ on the same product Gauss-Hermite grid, one extra hyperparameter p fitted as $\text{atanh}(p)$ with prior $\text{Normal}(0, 0.75)$ on the atanh scale (weakly informative: 95% prior mass within $|p| < 0.9$; motivation: the posterior-mean correlation -0.700 must be reachable without being presumed). Report its σ_{pop} , p , and the 50/80/95th percentiles alongside the two existing variants.
- **Accept:** summary regenerated; the LOPO comparison quoted under product-grid integration; the correlated fit either converges with usable SEs or is reported as not identified (also an acceptable outcome — 43 drivers is thin for a correlation).
- **Decision rule (pre-stated at R.1):** if the correlated variant moves the 80th percentile by more than the hierarchical variant's current CI half-width (~ 250 deficit units), escalate to review before any percentile is quoted downstream; otherwise the hierarchical variant stands as primary with the correlated fit reported as a robustness line.
- **Stop if:** the correlated variant's optimizer or Hessian fails after two honest attempts — report, keep the hierarchical variant primary, carry the query.

Card B.3.v2 — the LTAP field (re-scoped 2026-08-28)

(Replaces the B.3 sketch below for planning purposes; that card's construction advice still stands. Re-scoped because the LTAP data turns out to be the richest transfer target, not the hardest-to-justify one.)

- **Why it moved up:** 3 096 Random trials, the same 43 participants, **18 well-filled cells** (9 PET levels $\times$ 2 approach speeds, 172 trials each) and an intervention range of 0.110 – 0.907 — a wider dynamic range than the cyclist overtake (0.140 – 0.686) and comparable to the cut-in. The cyclist overtake, by contrast, has one condition (1.5 m) that is flat across all five timepoints, so it carries roughly two informative criticality levels.
- **What it uniquely offers:** the two-speed axis is the identifying variation the roadmap §0b wants for the *secondary* model — `t_react` scales with `v_ego` and `a_0v`, min with stopping distance, so 50 vs 70 km/h at matched PET separates them. Note the direction, which is worth understanding before modelling: intervention is **lower** at 70 km/h at every PET level (e.g. PET2: 0.605 at 50, 0.360 at 70).
- **What it lacks:** no timepoint — there is no truncation series, so LTAP yields a criticality $\times$ speed surface and not a criticality $\times$ time surface. Cross-scenario comparison must therefore be on criticality, or per-driver (see below).
- **Build:** the construction note first, per the original B.3 card — crossing geometry, arrival-time separation at the conflict zone, co-occupancy gating. The cut-in x/y conventions are meaningless here.
- **Cheap partial comparison, available now and already run:**
`replication/czb/cross_scenario_consistency.py` compares all four scenarios with **no field at all**, using the fact that every participant saw all four. Per-driver criticality-adjusted propensity correlates $+0.50$ to $+0.74$ across scenario pairs, which is 0.53 – 0.78 of the split-half reliability ceiling (mean 0.69).

Card A.1 — synthetic-recovery harness

- **Goal:** prove the stage-1 estimator recovers known parameters before touching real data.
- **Build:** `replication/czb/fit_recovery.py`. Simulate the Random cut-in design (43 participants $\times$ 18 cells $\times$ 4 repetitions) from the generative model in `docs/czb_fitting_plan.md` §2 stage 1: $c_i \sim \text{LogNormal}(\mu, \sigma)$, lapse floor b (both variants), $P = b + (1-b) \cdot \Phi((x - c_i)/\sigma_{\text{resp}})$ with x = the cell's `deficit_max` from `comfortzone.czb_data.random_cutin_trials()`. Fit by MAP in torch (L-BFGS) with Laplace SEs; also fit the `a_allowed` parameterization (threshold on `a_req_max`).
- **Accept:** over 20 simulated datasets, population μ and σ recovered within 2 Laplace SEs in ≥ 18 ; per-driver `c_i` shrinkage visible (posterior means closer to truth than raw per-driver MLEs in RMSE). Runtime under 10 minutes total.
- **Pitfalls:** identifiability of $(c, \sigma_{\text{resp}})$ scale — fix σ_{resp} 's prior scale from the pilot (≈ 2 200 deficit units); lapse and threshold trade off at C1 — check the recovery specifically at b .
- **Stop if:** recovery fails after two honest attempts — report the failure mode, do not loosen the acceptance criterion.

Card A.2 — stage-1 hierarchical fit, both bias variants

- **Goal:** the population distribution of the boundary level, and the bias-model decision.
- **Build:** `replication/czb/fit_stage1.py`, reusing A.1's model code on the real trials. Fit all four combinations: {deficit axis, `a_req` axis} × {group-level b, hierarchical `b_i`}. The ordered braking-expectation response enters as two nested thresholds (fitting plan §2); both levels free per roadmap §0b.
- **Report** (`replication/czb/out/stage1_summary.md`): population μ , σ with SEs; percentile table (50–95 in steps of 5) of the population distribution with CIs; held-out log-likelihood (leave-one-participant-out, 43 folds); posterior predictive $P(\text{intervene})$ per C1 cell against observed; fitted comfort and dread levels translated to `a_allowed` units and to steady-following THW at 20 m/s via `comfortzone.field.critical_thw`.
- **Accept:** all four fits converge; the summary table renders; LOPO likelihood distinguishes the bias variants or states that it cannot.
- **Stop if:** the C1 cells misfit under both bias variants (systematic sign pattern across criticalities) — that is the anticipation-leak question and needs review.

Card A.3 — the accumulator layer (review gate on its verdict)

[Verdict recorded 2026-08-27 at review gate R.1: FAIL, final and quotable. The gate repaired two diagnosed misspecifications (v2: noise gated at manoeuvre onset; v3: a free trial-level threshold spread, the analogue of stage 1's σ_{resp}), verified v2's degenerate optimum was global (`fit_stage2.py --scan`), and pre-committed to v3 as the last iteration. v3: in-sample 0.178 vs the static probit's 0.125; held-out 0.267 against the $\leq 0.11 / > 0.13$ rule. Structural cause in `out/stage2_summary.md`. Do not re-run this card; the accumulator question is closed for this stimulus set.]

- **Goal:** the pre-registered test: does time-integration close the within-scenario gap?
- **Build:** `replication/czb/fit_stage2.py`. Evidence = the per-frame deficit series (not the running max) from `stimulus_field`; linear accumulator with gain and noise, threshold from A.2's `c_i`; $P(\text{crossed by clip end})$ per trial via the closed-form Gaussian first-passage approximation or 200-path simulation per cell (cells share stimuli, so simulate per cell, not per trial). Motor latency fixed at 0.25 s; sensitivity refits at 0.15 and 0.35 s.
- **Accept / decision rule (pre-stated):** held-out (leave-one-criticality-out) RMSE on the 18-cell surface ≤ 0.11 counts as closing the gap (the 2D state rule's 0.120 and the design ceiling's ~ 0.10 bracket the target); > 0.13 counts as failure and is reported as evidence against the framework per the assessment.
- **Stop if:** the fitted gain wants to be so large the accumulator degenerates to the static threshold — report, since that itself answers the question.

Card A.4 — button-side validation

[Re-run 2026-08-27 at review gate R.1 under the final (v3) accumulator: still FAIL (worst checkpoint discrepancy 0.310), attributable to the accumulator's recorded A.3 inadequacy. What survives: the paradigm effect's direction (earlier pressing in Button) is confirmed; the level-versus-rate form is NOT distinguished (both shifts reach RMSE 0.163), so handbook ch. 11 decision 3 stays open. Do not re-run without a new response model.]

- **Goal:** predict the press-time densities from the Random-fitted model with one paradigm shift.
- **Build:** `replication/czb/validate_button.py` on `czb_data.button_cutin_trials()`. Fit δ once as a threshold shift and once as a gain shift; compare predicted vs observed press-time distributions per criticality (QQ plots to figures/, one page) and report which shape fits.
- **Accept:** the documented cross-paradigm excess (+0.26 at TTC6) is reproduced by the shifted model within 0.05 at the fixed-clip checkpoints.

Cards B.1–B.3 — transfer-scenario field constructions

Template for each: a loader in `src/comfortzone/<scenario>.py` following `cutin.py`'s structure (role assignment appropriate to the scenario — **by instructed vehicle, not lateral span**), a field check script in `replication/czb/` producing a per-criticality deficit figure, property tests, and the PS-versus-field correlation (the `ps_vs_field.py` pattern) as the acceptance readout.

- **B.1 cyclist overtake** (first): ego closes on a cyclist in lane; rear-end field nearly unchanged; criticality = lateral clearance 0.5/1/1.5 m, so the lateral term matters — use the actual clearance in `dy`, widths from the trace (cyclist ≈ 0.6 m). Accept: deficit ordered by clearance at C2–C5; cell-level PS Spearman ρ reported. *[Executed 2026-08-28; full account in `docs/overtake_construction_note.md`. The loader is `src/comfortzone/overtake.py`, validated against the study's own clearance labels (0.506 / 1.004 / 1.501 m against 0.5 / 1 / 1.5). Three corrections to this*

card: roles must come from the instructed vehicle because the cut-in's lateral-span rule is inverted here (the ego moves, not the target); the lateral coordinate is `Location_Y — Offset` inverts the criticality ordering; and the cyclist width is in the trace (0.582 m), not assumed. The card's premise did not hold: the field orders the 15 cells at Spearman +0.402 against the clearance label's -0.833, because `p_lane` sits at 0.843–1.000 all through the window and cannot see the manipulated variable. The scenario's criticality is a lateral comfort margin, not a collision geometry. Adding a lateral-clearance comfort term is a preference-function change and is left as a review decision.]

- **B.2 truck overtake / truck lateral:** ego passes a truck with clearance variants; the conflict is lateral, so `P_lane` runs on the *lateral closing* induced by the truck's drift (`truck_lateral_movement` traces) and the clearance term carries the statics. Accept: same criteria; also record whether the zero-deficit issue appears.
- **B.3 LTAP:** crossing geometry. Build arrival-time separation at the conflict zone (the intersection scenario's construction): compute each vehicle's time to the crossing-region entry/exit from the traces, and gate the collision/safety terms by predicted co-occupancy (the `P_lane` idea with lateral overlap replaced by zone co-occupancy). This card has real design content — **write the construction note first (1 page, the lane-entry-note pattern) and get it reviewed before coding.**
- **Stop for all three if:** a trace family's geometry does not fit the template (e.g. missing vehicle, broken dimensions beyond the documented truck case).

Card B.4 — the transfer test (review gate; do not run before A.3 and B.1–B.3)

[Adjusted 2026-08-27 at review gate R.1: A.3's FAIL removes the accumulator from this card. The transferred model is the stage-1 static threshold, hierarchical-lapse variant (population median 5 352, σ 0.209), everything frozen. The three-way comparison (one-scalar field vs per-scenario 2D rules vs elliptical joint) is unchanged.]

[Adjusted again 2026-08-28, on evidence from the B.1 pilot transfer (`docs/overtake_construction_note.md §6`). The primary analysis should freeze the boundary and free each scenario's lapse, not freeze everything. The pre-onset intervention rate is 0.081 in the cut-in against 0.198 in the cyclist overtake, so a no-shift primary starts every transfer cell ~0.12 low for reasons unrelated to the boundary: frozen transfer scores RMSE 0.176 against chance 0.158, while freeing only the lapse gives 0.129 — past chance and within 0.016 of the full-refit ceiling. The lapse is identified by that scenario's C1 cells alone, which end where the field is zero by construction and therefore carry no boundary information; the roadmap's secondary "one uniform level shift per scenario" does absorb the boundary and should stay secondary. Also: score against the field-independent ceiling in `out/cross_scenario_consistency.md` — only ~69% of the reliable per-driver signal is shared between scenarios, so a perfect one-scalar model cannot reach 100%.]

- Fit on cut-in only (A.2/A.3 configuration frozen); predict each transfer scenario's response surface with no refit. Primary: no per-scenario shift. Secondary: one instruction shift per scenario. Comparators on identical folds: per-scenario 2D state rules (1/TTC-analogue + the scenario's own second observable), and — once specified with Jonas — the elliptical joint on the same observables.
- Output: one table, per scenario × model: held-out corr and RMSE against (a) chance and (b) the per-scenario-refit ceiling. No verdict prose from the executing session; the verdict is a review task.

Card C — percentile sensitivity

- `replication/czb/percentile_sensitivity.py` from A.2's population distribution: implied trigger onset per stimulus for percentiles 50–95 (step 5) × `a_allowed` on the fitted comfort–dread range; report `d(onset)/d(5 percentile points)` in seconds against the percentile's own CI width. One figure, one table.

Card E.1 — closed-form versus sampled lane-entry expectation

- Monte Carlo the released binary gates under steering noise (heading random walk, $\sigma_w = \text{CUTIN_W_SD_MODEL}$, the model's own `dt`) along the 12 cut-in clips; compare the sampled conflict-probability with `lane_entry_weight`. Accept: max absolute difference reported per clip; agreement within ~0.15 through the response window validates the closed form, larger structured disagreement goes to review.

Cards E.2, E.3 and D

Deliberately not carded for cheap execution: E.2 (uncertainty term) and E.3 (policy-set field) involve model design, and D (safety audit) touches the causation runner's intervention injection. These get cards after their design notes exist.

4 Suggested session pattern

One card per session, in card order within a stage; start each session with "read handover.md, docs/czb_validation_roadmap.md, and docs/czb_work_orders.md, then execute card X and nothing else"; end each with the card's report, a commit whose message names the card, and one paragraph appended to the running log in replication/czb/out/worklog.md. Review sessions (stronger model) after A.2, A.3, and B.4 at minimum.

[Card order after gate R.1, 2026-08-27. Stage A is closed (A.1–A.4 done; the A.3/A.4 FAILs are recorded verdicts, not open work). The remaining cards, in the order a cheap-model session should take them:

- ¹ *B.1 — cyclist-overtake field construction (small, next).*
- ² *C — percentile sensitivity (small, cheap; uses A.2's decided hierarchical variant: median 5 352, σ 0.209).*
- ³ *A.2.v2 — regenerate stage 1 under the corrected validation code + the correlated-effects check (long-running; a good overnight companion to B.1 or C).*
- ⁴ *B.2 — truck field construction; B.3 — LTAP (write its construction note first and have it reviewed before coding, per the card).*
- ⁵ *B.4 — the transfer test (review gate; static stage-1 model per the R.1 adjustment).*

Queries in replication/czb/out/query_register.md: A.1.Q2 (a_req axis) stays a blocker for the truck check only; R.1.Q1–Q3 are for Jonas/review and block nothing above.]